

Question		Answer	Mark	Mark scheme	Additional guidance
21	(a)	$\frac{x+1}{3}$	M1	first step to change the subject of $y = 3x - 1$ or $x = 3y - 1$, eg $y + 1 = 3x$	
			A1	oe	
	(b)	Shown	M1	for method to find $fg(x)$, eg $fg(x) = 3(x^2 + 4) - 1$	
			M1	for method to find $gf(x)$, eg $gf(x) = (3x - 1)^2 + 4$	
			M1	(dep on previous two M marks) for setting up equation, eg $3(x^2 + 4) - 1 = 2[(3x - 1)^2 + 4]$	
			M1	(dep 2nd M1) for correct expansion of $(3x - 1)^2$ eg $9x^2 - 3x - 3x + 1$	
			C1	for $15x^2 - 12x - 1 = 0$ from correct working	